ALAN ROY

Full Stack Python Developer & Backend Engineer

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PROFESSIONAL SUMMARY

Results-driven **Full Stack Python Developer** with 1+ year of production experience architecting end-to-end web applications, asynchronous real-time services, and hardware-software IoT integrations. Proven track record of independently designing, scaling, and maintaining mission-critical enterprise platforms using **Python, Django, Django REST Framework, WebSockets (Daphne/ASGI)**, and **Redis**. Experienced in frontend development (Vanilla JS, TypeScript, React), database query optimization (PostgreSQL, SQLite), cryptographic workflows (PKCS#11/USB DSC signing), and offline-first Progressive Web Apps (PWA).

TECHNICAL SKILLS

Languages:	Python (Advanced), JavaScript (ES6+ / Vanilla JS), TypeScript, SQL, HTML5, CSS3
Backend & APIs:	Django, Django REST Framework (DRF), ASGI, Daphne, Django Channels, WebSockets, RESTful APIs
Databases & Caching:	PostgreSQL, SQLite, Redis (Session Management, In-Memory Caching, Pub/Sub), Indexing & Query Tuning
Frontend & UI:	React.js, Tailwind CSS, Vanilla CSS, DOM Manipulation, Leaflet.js, Google Maps Platform APIs
Architecture & Security:	Progressive Web Apps (Service Workers, Cache API), PKCS#11 Hardware Digital Signatures (PyHanko, TSA), IoT Sockets (ZKTeco Protocol), JWT Auth
Tools & Workflows:	Git, GitHub, Postman, WeasyPrint, Vite, AI-Augmented Software Development

PROFESSIONAL EXPERIENCE

Full Stack Developer (Python / Django) — Vadrida Technologies — Dec 2025 – Present | Kochi, Kerala

- **Enterprise Platform Architecture:** Solely architected and engineered the enterprise valuation ecosystem from ground up, automating verification workflows across 15+ banking formats (SBI, SIB, Muthoot, Axis) and reducing turnaround time by 75%.
- **Real-Time Asynchronous Engine:** Designed high-concurrency real-time subsystems using **Daphne (ASGI)** and **Django Channels (WebSockets)** for live presence tracking, dynamic dashboards, and instant updates across field and office teams.
- **Biometric IoT & Cryptographic Security:** Engineered hardware-level biometric device communication over TCP sockets (ZKTeco protocol) and built PKCS#11 USB DSC digital signature workflows with TSA RFC 3161 timestamping and SHA-256 verification.
- **Offline-First PWA & Building Rules:** Developed an offline-first survey PWA with Service Workers and GPS capture for field engineers; built automated engineering valuation logic for statutory building rules (KPBR/KMBR compliance & FSI).
- **Performance Optimization:** Optimized database queries and response times by over 60% by deploying in-memory **Redis caching**, fine-tuned indexing, and bulk record aggregation.

KEY PROJECTS

KSR Bait & Tackle — Retail ERP & E-Commerce Platform [Live Site: ksrbaitandtackle.in**]** — React.js, TypeScript, Python, DRF, PostgreSQL, Tailwind

- Engineered and deployed a full-featured commercial e-commerce platform and retail operations ERP on a custom live domain.
- Built modular product catalog filtering, dynamic cart state management, and streamlined checkout ordering workflows with React, TypeScript, and Tailwind.
- Architected scalable RESTful API services using **Django REST Framework** with JWT authentication, rate limiting, and relational schema migrations in PostgreSQL.

CurateDrop — Automated E-Commerce & Affiliate Store Engine — Python, Django, Vanilla JS, Web Scraping, WhatsApp Ordering

- Engineered an automated product ingestion pipeline extracting and normalizing Amazon product attributes, media, and specifications from arbitrary product URLs into structured store listings.
- Designed hybrid monetization workflows: routing affiliate inventory via tagged Amazon commission links and direct store products via WhatsApp message order confirmations.
- Implemented a secure delivery verification workflow featuring time-sensitive **Delivery OTP verification** and built a responsive mobile-first administrative management dashboard.

INTERNSHIPS & EDUCATION

Full Stack Python Intern — Logiccraft Technology

Engineered dynamic web modules; gained hands-on experience in MVT architecture, ORM queries, and RESTful API integration.

Full Stack Python Trainee — QSpiders Kochi

Intensive hands-on training in Python, Django, SQL database schema design, and Object-Oriented Programming (OOP).

Bachelor of Computer Applications (BCA) — Graduated 2025

Naipunnya School of Management

University of Kerala